
BRAVING THE STORM: STRENGTHENING THE PHILIPPINES’ DISASTER PREPAREDNESS AND RESILIENCE

CZARELLE F. LUCES

College of Law, West Visayas State University

The highest human right is the right to life. One has the expectation to have a healthful and sustainable environment, to live without any fear of destruction, and to have the capacity to resist any disastrous effects of inevitable natural calamities. This is supported by Section 16 of Article II of the 1987 Philippine Constitution which provides that *the State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.*¹

The Philippines is one of the world’s most disaster-prone countries. Located along the boundary of major tectonic plates and at the center of a typhoon belt, its islands are regularly affected by floods, typhoons, landslides, and earthquakes. The Philippine government has developed strong coping mechanisms over their long history of experience with disasters. Yet, significant gaps remain in natural disaster management capacities across different regions of the Philippines and surprisingly little data are available referencing local levels of disaster resilience and preparedness.

In a news report, The World Risk Index, a part of the annual risk report which calculates the disaster risk for 193 countries, showed that the Philippines has the highest disaster risk, with an index score of 46.82. The Philippines’ current ranking in the World Risk Index was a huge jump from its previous records. In 2018, the country ranked third on the list of countries with the highest disaster risk, with an index score of 26.70. In 2019, the country dropped to ninth after registering an index value of 20.69. In 2020, as the COVID-19 pandemic hit countries across the globe, the Philippines was still among the top 10 countries with the highest disaster risk while remaining in ninth place with the same index score. Last year, the Philippines ranked eighth with an index score of 21.39.²

Exposure

The report also found that the Philippines was among the top 10 countries with the highest exposure to disasters. The country ranked fourth with 39.99 exposure, trailing China, Japan, and Mexico. “Thus, exposure consists of the aspects of

¹ CONST. (1987), art II, §16 (Phil.).

² Baclig, C. E. *PH most disaster-prone country in the world—study* | *Inquirer News*. INQUIRER.net., (October 7, 2022), <https://newsinfo.inquirer.net/1676227/ph-most-disaster-prone-country-in-the-world-study>

hazardousness, which include the frequency and intensity of earthquakes, tsunamis, coastal and river floods, cyclones, droughts, and sea-level rise in an area (hazard zone) and populations (hazard object),” the report added.³

Natural disasters are natural hazards that result in great injury and damage. Natural hazards (such as extreme weather, wildfires, or earthquakes) may be inevitable, but disasters – with loss of life and high economic costs – are not.⁴ This is also referred to as natural hazards that are extreme sudden events caused by environmental factors such as storms, floods, droughts, and fires-now occurring with increasing severity, scope, and impact.

How Disaster Affects the Philippine Environment

For decades, the Philippine environment has been under assault from a fast-growing population and practices that have degraded the country’s air, land, and water. Despite laws requiring trees to be replanted, forests were clear cut and denuded areas abandoned, damaging watersheds. Consequently, this created and increased the desire to introduce sound environmental practices and various policies. The landmark statute that has been passed, which is the *Philippine Disaster Reduction and Management Act*,⁵ is however, underfunded and poorly implemented. These are successfully achieved when started in a community.

According to the Asian Disaster Reduction Center (ADRC), the Philippines is visited by an average of 20 typhoons each year, and most are destructive.⁶ This is supported by records of the Nationwide Operational Assessments & Hazards at the University of the Philippines (NOAH) Center that on average, disasters happen once a year and at most twice per year. But from 2017-2018, there were six in the span of one year.⁷ The impacts of typhoons can range from extensive property damage and economic loss to loss of human life. Record reveals that Typhoon Yolanda, which

³ Baclig, C. E., *PH most disaster-prone country in the world—study* | *Inquirer News*. INQUIRER.net., (October 7, 2022), <https://newsinfo.inquirer.net/1676227/ph-most-disaster-prone-country-in-the-world-study>

⁴ Scheunemann, C., *What is disaster risk reduction and why is it important?* Sphera., (June 24, 2024), <https://sphera.com/blog/what-is-disaster-risk-reduction-and-why-is-it-important/>

⁵ Philippine Disaster Reduction and Management Act (May 27, 2010) (Phil.), <https://www.officialgazette.gov.ph/2010/05/27/republic-act-no-10121/>.

⁶ Asian Disaster Reduction Center (ADRC). (2017), <https://www.adrc.asia/nationinformation.php?NationCode=608&Lang=en>

⁷ Novio, E., *Climate Change and Disasters in the Philippines* | Heinrich Böll Foundation | Southeast Asia Regional Office. Heinrich Böll Foundation | Southeast Asia Regional Office (January 21, 2022), <https://th.boell.org/en/2022/01/21/climate-disasters-philippines>

occurred in November 2013, was the most devastating typhoon in the Philippines in terms of the total value of damages. Damages caused by the typhoon reached about 95.48 billion Philippine pesos. With the international name Haiyan, this was also the costliest typhoon that has ever occurred in the country since 1990.⁸ This has been a great result of all the sudden and unexpected occurrence of disaster, which could have been prevented when everybody was prepared and equipped.

Disaster Legislation of the Philippines: How Effective is it?

With all these disastrous effects of calamities in the country, everybody seemed to be resilient and emotionally prone to every event. Everyone acts as if typhoons have already become part of our lives. But I guess, the word *resiliency* is now overused and should not be taken for granted. Life matters and that should be given the highest priority. Learning from its previous experiences in dealing with disasters, the Philippines has gradually shifted its approach from reactive to proactive risk management with a significant focus on preparedness and resilience building.

Known as the *writ of kalikasan*, or a ‘writ of nature’, this legal remedy that is unique to Philippine law provides for the protection of one’s right to “a balanced and healthy ecology in accordance with the rhythm and harmony of nature”, and is accorded as a constitutional right to the people of the Philippines. With this, the Philippine government has already implemented various legislation in combating and preventing harmful effects of disasters, particularly the program in *Disaster Risk and Reduction Management*. The bottom line is, how effective it is and what are the loopholes?

The Philippine Disaster Reduction and Management Act (R.A. No. 10121) or an act strengthening the Philippine disaster risk reduction and management system, providing for the national disaster risk reduction and management framework and institutionalizing the national disaster risk reduction and management plan, appropriating funds therefor and for other purposes,⁹ is an act of 2010 that was approved by previous president Gloria Macapagal-Arroyo.

⁸ Statista, Most disastrous typhoons Philippines 2011-2022, by value of damages, (January 4, 2024), <https://www.statista.com/statistics/1278568/philippines-most-devastating-typhoon-by-value-of-damages/#:~:text=Most%20disastrous%20typhoons%20Philippines%202011%2D2022%2C%20by%20value%20of%20damages&text=Typhoon%20Yolanda%2C%20w high%20occurred%20in,about%2095.48%20billion%20Philippine%20pesos>

⁹ Philippine Disaster Reduction and Management Act, Rep. Act No. 10121, (May 27, 2010) (Phil.), <https://www.officialgazette.gov.ph/2010/05/27/republic-act-no-10121/>

Its purpose primarily focused on implementing an efficient response regarding risks and disasters. It aims to give justice to the people's constitutional rights to life and property by developing disaster risk and management plans. The core values to which it adheres are good governance such as transparency and accountability within the context of environmental protection. It also provides assistance to those individuals and families affected by the disasters with the objective of normalizing and resuming social and economic activities.

Organization at the local government level is also discussed, which is the composition of the Local Disaster Risk Reduction and Management Council (LDRRMC) from the local executives to the representatives. Along with this are the offices that should be established at every province, city, municipality, and the barangay. The law also includes the mandate of integrating of Disaster Risk Reduction Education into the school curricula and Sangguniang Kabataan, which encourages community specifically the youth in participating into disaster, risk and management activities. This is where individual participation should come in, in order to successfully achieve this business. The Act shifted the policy environment and the way the country deals with disasters from mere response to preparedness. It provides a comprehensive, all-hazard, multi-sectoral, inter-agency, and community-based approach to disaster risk management through the formulation of the National Disaster Risk Management Framework. This is being formulated, developed, and implemented as the master plan that will provide the strategies, organization, tasks of concerned agencies and local government units, and other guidelines in dealing with disasters or emergencies. Through this plan, coherent, integrated, efficient, and responsive disaster risk management at all levels will hopefully be achieved.

It also recognizes local risk patterns and trends and decentralization of resources and responsibilities and thus the full participation of the Local Government Units (LGUs) and communities in governance. The approach tends to be 'response-oriented' or 'reactive.' This is evidenced by the widespread emphasis on post-disaster relief and short-term preparedness, such as forecasting and evacuation, rather than on mitigation and post-disaster support for economic recovery. Moreover, the Act mandates the establishment of a Disaster Risk Reduction and Management Office (DRRMO) in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in every barangay. This provides for the calamity fund to be used in support of disaster risk reduction or mitigation, prevention, and preparedness activities for the potential occurrence of disasters and not just for response, relief, and rehabilitation efforts.

Reflected in the United Nations Office for Disaster Risk Reduction (UNDRR), the Disaster Risk Reduction (DRR) report provides a summary of the DRR

progress the Philippines has achieved under all of the priorities.¹⁰ Recently in their 2023 report, it shows that at a global level, there have been significant advances in developing national DRR strategies. The number reported increased from 55 in 2015 to 125 in 2022 (127%). Training and capacity-building opportunities to monitor implementation contributed to this progress. At a regional level, major progress has also been witnessed. Regions adopted regional strategies at regional platforms, which played important roles in improving risk understanding and awareness, and accelerating DRR implementation. At a local level, the number almost doubled, from 51 in 2015 to 91 in 2022.¹¹

Challenges and Lapses

Disaster preparedness plans reduce future damages but may lack testing to assess their effectiveness in operation. The local governments were found highly vulnerable to tropical cyclones and floods while vulnerable to earthquakes, droughts, and landslides. They were partially prepared regardless of profile, but the coastal, middle-earning, most populated, having the least number of villages, and middle-sized had higher levels of preparedness. Those highly vulnerable to earthquakes and forest fires were prepared, yet only partially prepared to floods, storm surges, droughts, tropical cyclones, tornadoes, tsunamis and landslides. The diverse attitude of stakeholders, insufficient manpower, and poor database management were the major problems encountered in executing countermeasures. Appointing full-time disaster managers, developing a disaster information management system, and massive information drive, organizing village-based volunteers, integrating disaster management into formal education, and mandatory trainings for officials, preparing for a possible major volcanic eruption, and crafting a comprehensive plan against emerging emergencies may lead to a 360° preparedness.

Community Engagement Should be Enhanced

A person's safety during calamities is mainly dependent on their community's disaster preparedness. No matter how ready they are for all possible catastrophic scenarios, every bit of preparation can go down the drain if their community cannot respond effectively to such events. Community participation has been recognized as the additional element in disaster management necessary to reverse the worldwide

¹⁰ Disaster risk reduction in the Philippines: Status Report 2019, UNDRR.org., <https://www.undrr.org/publication/disaster-risk-reduction-philippines-status-report-2019>

¹¹ STATUS REPORT ON TARGET E 2023, UNDRR.org., <https://www.undrr.org/publication/status-report-target-e-2023>

trend of increasing frequency and loss from disasters, build a culture of safety and disaster resilient communities, and ensure sustainable development for all.

The above progress is encouraging and brings promises. But there are still opportunities for improvement. There is a pressing need for strengthening and empowering community-based disaster management (CBDM) as showcased by good practices in community-based preparedness.

Why disaster risk reduction is everyone's business? It involves life- your life. One must engage in community preparedness initiatives, such as neighborhood emergency response teams or local disaster preparedness programs. Collaborating with neighbors and local authorities allows one owner to share resources, information, and support during emergencies.

A study shows that community engagement is very effective. The early delivery of a warning that affects the alertness of the residents had been confirmed effective by most of the respondents. In a 2020 study on *the Effectiveness of Community-Based Disaster Management in Cabanatuan City*, results showed that the prevention and minimizing the impact of natural disasters made by the barangay were properly prepared and necessary actions were taken properly. In terms of conducting a short-term recovery period there is a sufficient supply of relief goods that helps the residents to recover faster and the rescue team conducts their job without further delay. Thus, effective training is needed to improve the performance of the rescue team and incentives might also be used in improving it.¹²

Also, in the 2018 study on *the Level of Effectiveness of Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in the Selected Municipalities in the 3rd Congressional District of Quezon*, the result of the study evidently confirmed that the BDRRMC are effective in performing their duties/functions and responsibilities in terms of setting direction, development, implementation and coordination, but based on the response of the stakeholders the BDRRMCs appeared to be mostly moderately effective specifically in terms of implementation and coordination. Furthermore, the BDRRMC still encountered different problems and challenges, which include insufficient training and capability building, road access, communication, awareness, politics, and the unavailability of Disaster Risk Reduction and Management supplies and equipment.¹³

¹² Santos, K. E., *Effectiveness of Community-Based Disaster Management in Cabanatuan City*. Research Gate., (2020), https://www.researchgate.net/publication/338856470_Effectiveness_of_Community-Based_Disaster_Management_in_Cabanatuan_City

¹³ Glindo, M. S. A., *Level of effectiveness of Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in the selected municipalities in the 3rd*

Utilizing the findings of these studies, one could propose a program enhancement for the Barangay Risk Reduction and Management Committee which was deemed very effective. Strengthening local institutional capacity for DRR should be the highest priority. Understanding the different capacities and constraints of local governments is critical. Local governments and authorities are often knowledgeable about the local needs and gaps, and their support in the operations, especially in the remote regions, is important to guarantee the actualization of intended aspirations for future resilience and sustainability not only in recovery, but also in preparedness and mitigation. There is a need to develop a harmonized and comprehensive planning process with a clear understanding of the linkages at the local level.

Disaster risk reduction is everyone's business. Every individual is a material particle in pushing through this significant objective towards achieving a healthy environment. Participating in the various advocacies of environmental law will create in every individual this altruistic feeling, which would be very beneficial to its country. Recognizing the constitutional guarantee of protecting and advancing the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature, would lead us all in a most prepared community. Implementing environmental legislations will only remain dormant and ineffective unless all of these loopholes and gaps are fully implemented with proper focus on gray areas. National government must start from the localities; community engagement should be enhanced.

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